Summary of this week

- Congratulations for making it to this point!
- You have now learned all the primary theoretical components that are required to derive and understand particle filters.
- This week, you learned how to implement the basic SIS algorithm in Octave.
- Then, you learned that it doesn't always perform well, due to particle degeneracy.
- But, you then learned how to perform resampling to correct that problem, and implement the improved method in Octave code as well.
- The final SIS-with-resampling algorithm gave near-optimal results but still beats the "curse of dimensionality."

Where to from here?

- This section on particle filters is only an introduction:
  - There is a lot of literature on choosing importance densities, on finding MAP estimators, etc.
  - Prof. McNames' course covers some of these more advanced topics, if you are interested.[a]
  - The material we have covered here should prepare you for his lectures and help you read and interpret the literature.
- We now take a different direction and look at navigation applications of Kalman filters.

[a] Recall, his course "State Space Tracking" can be found on YouTube at: `https://www.youtube.com/@JamesMcNames/videos`

Credits

Credits for photos in this lesson

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